prepare_sjsdm_cross_validation_fold.R

August 10, 2026

Contents

prepare_sjsdm_cross_validation_fold 1

Index **3**

`prepare_sjsdm_cross_validation_fold` *Prepare One sjSDM Cross-Validation Fold*

Description

Converts location-assignment row indices to sample identifiers, rebuilds fold-local spatial predictors when needed, and prepares train/test model inputs with training-only response filtering and predictor scaling.

Usage

```
prepare_sjsdm_cross_validation_fold(
  data_community_matrix = NULL,
  data_abiotic_wide = NULL,
  data_coords_projected = NULL,
  data_sample_ids = NULL,
  train_indices = NULL,
  test_indices = NULL,
  config_model_fitting = NULL,
  config_data_processing = NULL,
  repeat_id = NULL,
  fold_id = NULL,
  compute_spatial_function = compute_spatial_mev,
  compute_spatiotemporal_function = compute_spatiotemporal_mev,
  interpolate_spatial_function = interpolate_mev_to_grid,
  interpolate_spatiotemporal_function = interpolate_st_mev_to_grid
)
```

Arguments

`data_community_matrix`
: Raw community matrix with sample identifiers as row names.

`data_abiotic_wide`
: Wide abiotic data frame with 'dataset_name', 'age', and predictors.

`data_coords_projected`
: Projected coordinates used for fold-local MEM construction.

`data_sample_ids`
: Sample metadata containing 'dataset_name' and 'age'. Optional 'sample_id', 'row_index', and 'location_id' columns are derived when absent.

`train_indices`, `test_indices`
: Integer row indices defining the fold training and held-out samples.

`config_model_fitting`, `config_data_processing`
: Active project configuration lists.

`repeat_id`, `fold_id`
: Fold identifiers. They are accepted for runner compatibility and retained by downstream diagnostics.

`compute_spatial_function`, `compute_spatiotemporal_function`
: Functions used by [prepare_fold_spatial_predictors()] to build MEMs.

`interpolate_spatial_function`, `interpolate_spatiotemporal_function`
: Functions used by [prepare_fold_spatial_predictors()] to project MEMs.

Details

This helper is the production adapter between location-level CV assignment tables and the generic fold-local preparation functions shared with decomposition diagnostics.

Value

Named list returned by [prepare_model_fold_input()], with optional 'data_spatial_diagnostics' when spatial predictors are enabled.

Index

prepare_sjsdm_cross_validation_fold, 1